Table 7.5: Different types of inclusion bodies in bacteria

Type of inclusion bodies	Example of organisms possessing	Significance
Polyhydroxybutyrate (PHB)	*Bacillus megaterium*	Reserve of Carbon and energy sources. Sudan dye is used to observe lipid inclusions
Polyphosphate (volutin granules) or metachromatic granules	*Corynebacterium diphtheriae*	Reserve of phosphate
Sulphur globules	Phototrophic bacteria Like purple and green Sulphur bacteria Example: *Thiobacillus*	Elemental sulphur, reserve of electrons in phototrophs. Reserve of energy source in lithotrophs
Gas vesicles	Aquatic bacteria, *Cyanobacterium*	They are protein shells filled with gases. They provide buoyancy and keep the cells floating in vertical water column
Parasporal crystals	Genus *Bacillus*	It is a proteinaceous compound, It is toxic to certain insects
Magnetosomes	*Aquaspirillum magnetotacticum*	They are like intracellular chains of magnetite particles. They help the bacteria to swim to nutrient rich sediments. It protects the cell against H_2O_2 accumulation
Carboxysomes	Photosynthetic Bacteria, cyanobacteria Autotrophic bacteria	They contain the enzyme Ribulose 1-5 bisphosphate carboxylase which is involved in Carbon dioxide fixation during photosynthesis
Phycobilisomes or cyanophycin Granules	Cyanobacteria	They have a long polypeptide with equal proportion of Arginine and Aspartic acid. They store Nitrogen
Chlorosomes	Green bacteria	They contain bacteriocholorophyll pigments which are involved in bacterial photosynthesis

HOTS

Why are endospores so difficult to destory?

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